

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 6

5 Data Science (djw1005)

In order to judge whether early prehistoric humans had some form of religious belief, archaeologists have proposed studying the alignment of graves in a burial site. The thinking is that if the graves all point in the same direction then the society must have had some form of supernatural belief which led them to align the graves in this way. However, we wouldn't expect perfect alignment: the skeletal remains might be at an angle to the original grave; or there might be a mix of aligned and non-aligned graves from different waves of occupants.

You have been asked by an archaeologist friend to conduct a hypothesis test of whether a collection of graves shows alignment. The dataset is a list $[x_1, \dots, x_n]$ of angles, $0 \leq x_i < 360$ where 0 denotes North, 90 denotes East, and so on.

- hypothesis testing (a) What is meant by a “null hypothesis”? What null hypothesis do you propose? [5 marks]
- hypothesis testing (b) What is meant by a “test statistic”? What test statistic do you propose? Explain how your test statistic deals with the fact that 359.999 and 0 are very similar angles. [7 marks]
- hypothesis testing (c) What is meant by a “one-sided” or “two-sided” test? Which will you use? Explain your reasoning. [5 marks]
- hypothesis testing (d) Define “ p -value”. Explain how to compute the p -value for your test. [3 marks]

Answer:

— *Solution notes* —

Each question has a bookwork component worth two marks, and a creative modelling component for the rest of the marks.

Part (a). A null hypothesis (H_0) is a probability model for the data. We take it as true by default, and we look for evidence that it might not be true. In this case, the obvious null hypothesis is that the datapoints are independent $U[0, 360]$. [1 mark for ‘independent’, 1 mark for a distribution, 1 mark for getting null and alternate hypotheses the right way round.]

Parts (b) and (c). A test statistic $t(x)$ is any value that can be computed from the dataset x (i.e. which doesn’t involve any unknown parameters). We need to be able to work out the distribution of $t(X)$ for a random dataset X generated under H_0 , and we should be able to say how it is likely to differ if H_0 is false. If it tends to be larger when H_0 is false then we’ll reject H_0 for large values of $t(x)$; this is called one-sided. If it tends to be smaller when H_0 is false then we’ll reject H_0 for small values of $t(x)$; this is also called one-sided. If it may be either larger or smaller when H_0 is false then we’ll reject H_0 for either small or large values of $t(x)$; this is called two-sided. Some options... [2 marks for a valid statistic, 3 marks for an explanation of $359.99 \approx 0$, 3 marks for explaining its sidedness.]

- Naively compute $n^{-1} \sum_i (x_i - \bar{x})^2$ where $\bar{x}$ is the sample mean. If H_0 is false and the graves are aligned we’d expect this to be smaller. This statistic doesn’t account for $359.99 \approx 0$; if the graves are aligned around North then this test is unlikely to detect it. [This answer doesn’t get full marks for the explanation of $359.99 \approx 0$.]
- Split the data into bins, and count the number of observations y_m in each bin m . Measure $\sum_m (y_m - \bar{y})^2$ where $\bar{y}$ is the mean count. If H_0 is false and the graves are aligned we’d expect this to be larger. This statistic is insensitive to the $359.99 \approx 0$ issue.
- Let $t(x) = \min_{\theta} n^{-1} \sum_i \min(|x_i - \theta|, |x_i - 360 - \theta|)^2$. This is a very cunning way to get a uniformly good answer whatever the alignment. It’s like the first answer but it uses $\min_{\theta}$ as a proxy for $\bar{x}$ (note that $\bar{x}$ solves $\min_{\theta} \sum_i (x_i - \theta)^2$). It deals with $359.99 \approx 0$ by using the smaller error going either clockwise or anticlockwise.
- Formally, any statistic at all is a legitimate test statistic, since H_0 specifies completely the distribution of the data and we can always compute the distribution of $t(X)$! Some statistics are practically useless since they have poor ability to detect deviations from H_0 , but they’re still technically correct. Formally, we’re entitled to use whichever we like out of one-sided and two-sided; again, the bad choice will have poor power but is technically correct. Any student with the sophistication to make this argument deserves full marks!

Part (d). The p -value is the probability of seeing a test statistic at least as extreme as what we actually saw, assuming H_0 is true. The student should give the appropriate formula given the sidedness they declared in part (c). For example, for the first option, where we reject H_0 if the test statistic is small, $p = \mathbb{P}(t(X) \leq t(x))$ with X defined as above. In practice we can compute this by simulating many random datasets X and counting the fraction for which $t(X) \leq t(x)$.
